some particle species like the photon and π^0 are their own antiparticles; for these the field adjoints are proportional to the fields.

We now proceed to move all annihilation operators to the right in Eq. (6.1.1), repeatedly using for this purpose the commutation or anti-commutation relations:

$$a(\mathbf{p}\ \sigma\ n)a^\dagger(\mathbf{p}'\sigma'n') = \pm a^\dagger(\mathbf{p}'\sigma'n')a(\mathbf{p}\ \sigma\ n) + \delta^3(\mathbf{p}' - \mathbf{p})\delta_{\sigma'\sigma}\delta_{n'n} \tag{6.1.4}$$

$$a(\mathbf{p}\ \sigma\ n)a(\mathbf{p}'\sigma'n') = \pm a(\mathbf{p}'\sigma'n')a(\mathbf{p}\ \sigma\ n) \tag{6.1.5}$$

$$a^\dagger(\mathbf{p}\ \sigma\ n)a^\dagger(\mathbf{p}'\sigma'n') = \pm a^\dagger(\mathbf{p}'\sigma'n')a^\dagger(\mathbf{p}\ \sigma\ n) \tag{6.1.6}$$

(and likewise for antiparticles), the $\pm$ sign on the right being $-$ if both particles n, n' are fermions, and $+$ if either or both are bosons. Whenever an annihilation operator appears on the extreme right (or a creation operator on the extreme left), the corresponding contribution to Eq. (6.1.1) vanishes, because these operators annihilate the vacuum state:

$$a(\mathbf{p}\ \sigma\ n)\Phi_0 = 0\,, \tag{6.1.7}$$

$$\Phi_0^\dagger a^\dagger(\mathbf{p}\ \sigma\ n) = 0. \tag{6.1.8}$$

The remaining contributions to Eq. (6.1.1) are those arising from the delta function terms on the right-hand side of Eq. (6.1.4), with every creation and annihilation operator in the initial or final states or in the interaction Hamiltonian density paired in this way with some other annihilation or creation operator.

In this way, the contribution to Eq. (6.1.1) of a given order in each of the terms $\mathscr{H}_i$ in the polynomial $\mathscr{H}(\psi(x), \psi^\dagger(x))$ is given by a sum, over all ways of pairing creation and annihilation operators,[2] of the integrals of products of factors, as follows:

(a) Pairing of a final particle having quantum numbers $\mathbf{p}', \sigma', n'$ with a field adjoint $\psi_\ell^\dagger(x)$ in $\mathscr{H}_i(x)$ yields a factor

$$\left[a(\mathbf{p}'\ \sigma'n'),\ \psi_\ell^\dagger(x)\right]_\mp = (2\pi)^{-3/2}e^{-ip'\cdot x}u_\ell^*(\mathbf{p}'\sigma'n')\,. \tag{6.1.9}$$

(b) Pairing of a final antiparticle having quantum numbers $\mathbf{p}', \sigma', n'^c$ with a field $\psi_\ell(x)$ in $\mathscr{H}_i(x)$ yields a factor

$$\left[a(\mathbf{p}'\sigma'\ n'^c), \psi_\ell(x)\right]_\mp = (2\pi)^{-3/2}e^{-ip'\cdot x}v_\ell(\mathbf{p}'\sigma'\ n')\,. \tag{6.1.10}$$

(c) Pairing of an initial particle having quantum numbers $\mathbf{p}, \sigma, n$ with a field $\psi_\ell(x)$ in $\mathscr{H}_i(x)$ yields a factor

$$\left[\psi_\ell(x), a^\dagger(\mathbf{p}\sigma n)\right]_\mp = (2\pi)^{-3/2}e^{ip\cdot x}u_\ell(\mathbf{p}\sigma n)\,. \tag{6.1.11}$$